



Hypothesis Testing

Key terms

hypothesis testing, null hypothesis, alternative hypothesis, statistical significance, critical value, critical region, one-tailed, two-tailed

KSBs: K13, S11, S52, B1

6.1 Hypothesis Testing

Hypothesis testing is the process of applying statistical tests to experimental situations. An intervention, which it is suggested might cause a , is compared to the state without intervention - and statistical analysis can tell us whether there is a

Definition 6.1 - Null Hypothesis and Alternative hypothesis

A **hypothesis** is a statement about the effect of an intervention on a situation, which you wish to test in order to

The **null hypothesis**, denoted H_0 , is the statement encapsulating the situation in which there is
In the absence of sufficient evidence, we assume the null hypothesis
is true

The **alternative hypothesis**, denoted H_1 , is the statement we wish to test, and describes a

Many different types of statistical test can be used, and we can compare many different things, including whether statistics about a set of data what the distribution of the data looks like, or whether a sample a given distribution.

Example 6.1.

Suppose a company is considering putting a new type of coating on bearings that it produces. The mean wear life with the current coating is known to be The company would not want to make any (costly) changes unless evidence strongly suggested

The hypothesis we would seek to test in this case would be that

If we denote the new mean wear life by μ_1 , we would write

Exercise 6.1.

For each of the experiments described below, state the null hypothesis and the alternative hypothesis.

- A runner wishes to improve her 100m time, which is normally distributed with mean μ_0 . She trains intensively for a week, and measures her time for running 100m over 5 days.
- A group of 46 individuals with clinical depression were randomly assigned to either treatment as usual, or sessions playing a computer game. Participants performed a neuropsychological assessment which produced a score. The scores were then compared between the groups. Neuropsychological assessment scores are normally distributed, and higher scores on the test correspond to better mental health.
- A football fan wishes to determine if there is a biased pattern in the times at which yellow cards are given out during a match, or whether they follow a normal distribution. They watch all the games in a Premier League season and record the number of yellow cards given out during 10 minute intervals of each match.
- A computer gamer achieves a score of 40,000 points on a difficult game, which is a new world record. The scores for the game are normally distributed, and the adjudicator for the world record wishes to determine whether or not the player was cheating.

6.2 Experimental design

When we design experiments, we usually have a few common features.

- An independent variable, which you can control the value of; this could be a
- A dependent variable, which we can measure
- A sample of data to perform the test on. This sample

In order to make sure the effect you're measuring is due to the variable you're changing, it's often useful to apply the change to a _____ and leave the rest unchanged - called a **control group**. This allows you to measure _____ more clearly.

It's also important, if possible, only to alter _____ otherwise, any measured change in the results could be due to any of the things you've changed, with no way to tell.

It may be the case that **you aren't in control of the exact parameters of the experiment**: you may be performing tests on _____ or you may not have the ability to collect a _____ or establish _____. If you can obtain information about the source of the data and how it was collected, you can take this into account when analysing the data.

6.3 Statistical Significance

If we measure a change following the intervention, it isn't necessarily the case that this change was _____ - natural variation in the variable could coincide with the intervention, in a way that might

If data is drawn from an underlying _____ we would expect the values to vary

- with most values but some rarer examples will be

We use the concept of **statistical significance** to measure the difference between a and something which could be considered an

Definition 6.2 - p -value

The **p -value** is the probability of obtaining test results observed, under the assumption that means that such an extreme observed outcome would be

as the result actually
A very small p -value
under the null hypothesis.

Example 6.2.

Muriel Bristol claims she can detect whether a cup of tea has been made by
by tasting it. Her colleague prepares had milk poured
first, and the other

There are ways to choose 4 objects from a set of 8, so possible ways the cups could have been arranged.
Muriel and so the probability of this happening by chance is

In this experiment, the null hypothesis is that and has no way to tell if
the milk was poured in first. Since this result was to have occurred by chance, we can
and promote the alternative hypothesis - that Muriel has a special ability to
detect whether tea or milk was poured first.

Definition 6.3 - Significance Level

The **significance level** is the value we have decided the p -value needs to be in order
to reject the null hypothesis; conventionally, significance is set at

In our earlier example, so we can say that the result is

Note

5% is the standard significance level used across most of science and data analysis. There may be situations where other values are used, for specific types of scientific experiment - but generally 0.05 is standard.

Definition 6.4 - Test statistic

A **test statistic** is a single value obtained from a dataset in order to

Unlike a test statistics do not necessarily tell us much about the data, but they are instead used to test for significance.

In the examples we'll be looking at today which involve the normal distribution, the test statistic is the and this type of testing is called a For other types of hypothesis test we will see later, the test statistic is calculated differently and has a different name.

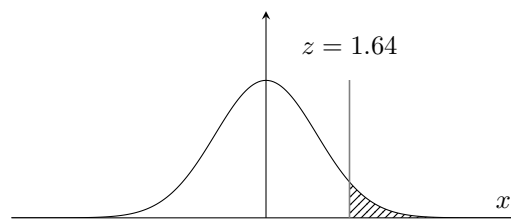
Definition 6.5 - Critical Value/Critical Region

For the significance level we have chosen, we can calculate the **critical value**, which is the value of the test statistic for which the p -value is

The **critical region** is the range of values the test statistic can take for which the p -value is so we can

The critical value for a p -value of 5% will be the value of the test statistic for which the probability is (or, the probability of it not happening is exactly).

For the normal distribution, by looking it up in the table, we see a probability of 95% corresponds to a z -score of , and anything to the right of this (with a z -score) will be in the critical region.



If we are seeking the value of the test statistic to be we call this the of the distribution (the right-hand end of the diagram), and if we seek a value we examine the (on the left of the diagram).

Example 6.3.

A factory makes widgets whose weights follow the distribution On visiting the factory, you find a stray widget on the floor, and want to know where it has come from. It may have either or have been who were in today from the Large Widget Factory down the road, whose widgets are

H_0 :

H_1 :

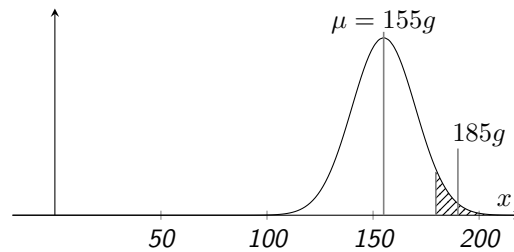
The widget is weighed, and it is found to have a weight of

The z -score (test statistic) for this value is

Looking this up in the table gives a p -value of therefore we might choose to the widget was from the other factory.

This is and assume

We can also note that a z -score of is greater than the critical value of 1.64, and



For this distribution, the weight corresponding to the critical value can be found by substituting back into the z -score calculation:

So if the widget's weight is greater than _____ we know we can

Note

It is important to remember that the probability we are calculating is not the probability that (e.g. in this case) the widget is from the production line; it is the probability that **if it were** (if the null hypothesis is true), **we would see a result this extreme**.

6.3.1 Testing the distribution

As well as testing a specific value for whether it plausibly comes from a set with a given distribution, we can test whether the _____ given an observation you know is from the set. This uses the

Example 6.4.

A farmer has been monitoring the amount of lead in one of her fields next to a factory over many years. Over time, she has found the mean density of lead to be _____ with standard deviation _____. Let X be a variable representing the amount of lead contained in the field. We have

After the factory closed, the farmer took _____ randomly selected from different parts of the field, and found the mean density of lead to be _____. We assume that the amount of lead follows a normal distribution with standard deviation 25mg/kg.

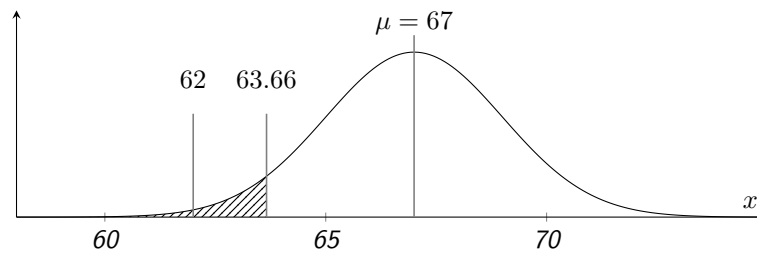
Since we are measuring a change in the mean, we can use the _____ which for samples of size 150 will be given by

We knew that the mean was _____ changed. It has been suggested that _____ to test.

so there's no reason to think it has _____ and that is what we would like

H_0 :

H_1 :



This is a lower tail test, so for a z -score to be in the critical region of $z < -1.645$ to reject the null hypothesis. For our sample, $z = -1.645$. The z -score for this value can be calculated as:

This is in the critical region, so we can reject H_0 and accept the hypothesis that the amount of lead has decreased.

A z -score of -1.645 corresponds to a probability of 0.05 which is

For a significance level of 5%, we can calculate the density corresponding to the critical value as:

This is the value to the right of the mean, and to find the equivalent value to the left we can calculate $z = 1.645$ and subtract this from 67:

If the density of the sample is below this value, we can

Exercise 6.2.

- A biologist studying elk measures the weights of elk in the park where they work over the course of a year, to make sure they are getting enough food. The biologist finds that based on a sample of the population, they have a mean weight of 171kg with a standard deviation of 47kg. Their weights are assumed to be normally distributed. An elk is observed in the park, and found to have a weight of 95kg. Test, using the 5% significance level, whether this elk is underweight.
- An ice cream van driver starts a round on a new estate. The mean daily income she receives on this estate during her first month is £46.70 with a standard deviation of £14.30. At the beginning of a second month, she records the daily income from this estate of a random sample of 7 days. The total income for those 7 days is £450. Test, using the 5% significance level, whether the mean daily income from this estate exceeded £46.70. Assume that the sample of daily incomes is from a normal distribution with standard deviation 14.3.

6.4 Two-tailed tests

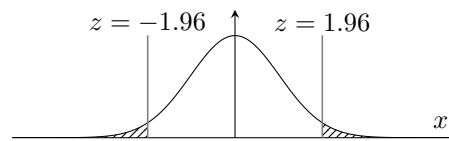
There may be occasions when we are looking for a value to be and not too far away in
either direction.

Definition 6.6 - One-tailed and two-tailed tests

We use the term **one-tailed test** to refer to a hypothesis test involving restriction
If we need the value to be - not
too far away from the mean in either direction - we use a **two-tailed test**, which studies the

For two-tailed tests, we need to calculate the critical values slightly differently - since we still want the result to have a we need to take into account that of
the distribution, so we instead draw the lines at and (as we did for the 95% confidence interval).

For a normal distribution, instead of 1.64, the critical value for the z -score will now be

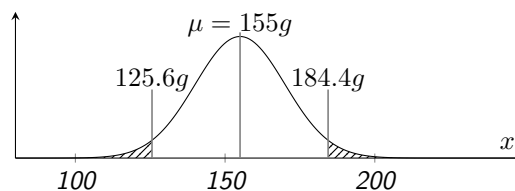


Example 6.5.

Back at the widget factory from Example 6.3, the following week, a delegation from the We Make A Range Of Sizes of Widgets But Often Drop Things Factory has been to visit. A spare widget is discovered on the floor with a weight of Did it fall off from the production line, or was it dropped by a visitor?

H_0 :

H_1 :



For a significance level of 5%, we can use the

and calculate

This is the maximum weight for a result not to be considered significant. In the other direction, we can calculate:

So if the widget weighs between

we can reject the alternative hypothesis and conclude

it was probably

Since our widget was 180g, we can

Exercise 6.3.

- (a) Research has shown that the mean recovery time for a cold is 18 days, with a standard deviation of 1.5 days, and follows a normal distribution. What are the upper and lower limits for the time taken to recover from a cold, to a significance level of 5%?
- (b) At a bakery, a machine is used to measure out quantities of dough for making loaves. For each batch, the mean amount of dough per load is set using a control on the machine. The weights of the resulting loaves are normally distributed with standard deviation 24 grams.

The control is set to produce a batch of loaves with a mean weight of 1000 grams. The weights, in grams, of a random sample of nine loaves from this batch had a mean of 970.1 grams. Using the 5% significance level, examine whether the mean weight of this batch of loaves is different from 1000 grams.

6.5 Structure of a Hypothesis Test

The outline below may be helpful when approaching a hypothesis testing situation.

- Define your variables
- Formulate the hypotheses (H_0 and H_1)
- Declare the test including:
 - which tail/how many tails
 - the type of test
 - the significance level and
 - what you need to carry out the test
- Calculate the test statistic
- Determine the critical region OR Determine the p -value
- Compare the test statistic with the critical region OR Compare the p -value with the significance level
- Determine whether the result is significant
- Decide whether to reject H_0 or not
- Determine the conclusion.

Exercise 6.4. For each of the situations described below, follow the list above and suggest what you would write for the first three bullet points.

- (a) A flock of goldfinches eat seeds from a birdfeeder, and the amount of seed each bird consumes is normally distributed. A bird visits the feeder while nobody is watching, and you wish to determine if it was a goldfinch, or one of the larger birds sometimes seen in the area.
- (b) The heights of 10-year-old boys are normally distributed. A researcher wants to determine if a new experimental drug some boys were given as babies has affected their height.

Later we will see more examples of types of test which can be applied to different situations.